

Mathematical Document Test

Enhanced LaTeX System

October 9, 2026

1 Basic Mathematics

Simple inline math: $a + b = c$ and $x^2 - 1 = (x - 1)(x + 1)$.

2 Fractions and Powers

Inline fractions $\frac{1}{2}$ and $\frac{a+b}{c+d}$.

Powers and subscripts: x^2 , y_i , z^{a+b} , $w_{i,j}$.

Nested structures $\frac{1+\frac{1}{x}}{\sqrt{1+x^2}}$ and $\left(\frac{a}{b}\right)^{n+1}$.

3 Display Mathematics

$$E = mc^2 \tag{1}$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \tag{2}$$

4 Greek Letters

$\alpha, \beta, \gamma, \delta, \epsilon$

$\pi, \sigma, \tau, \phi, \psi, \omega$

$\Gamma, \Delta, \Theta, \Lambda, \Pi, \Sigma$

5 Mathematical Operators

$$\sum_{i=1}^n i^2, \prod_{i=1}^n i, \int_0^1 x dx$$

$$\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$$

$$\sum_{k=0}^n \binom{n}{k} x^k y^{n-k} = (x+y)^n \quad (3)$$

$$\int_{-\infty}^{\infty} e^{-x^2} dx = \sqrt{\pi} \quad (4)$$

6 Relations and Decorations

Ordered relations $a < b \leq c \neq d \geq e > f$ and quantified statements $\forall \epsilon > 0$, $\exists \delta > 0$.

$$\bar{z} = a - bi, \quad \vec{v} = \hat{i} + \hat{j} \quad (5)$$

$$\left\langle \frac{u}{\|u\|}, \frac{v}{\|v\|} \right\rangle = \frac{u \cdot v}{\|u\| \|v\|} \quad (6)$$

7 Delimited Groups and Functions

Trigonometric and logarithmic functions $\sin \theta$, $\cos \theta$, $\tan \theta$, $\log x$, $\ln x$, and $\exp(x)$.

$$\left[\sum_{r=0}^m \frac{(-1)^r}{r!} \right]^2 = \left(\frac{1 - e^{-m}}{1 + e^{-m}} \right) \quad (7)$$

$$\left\lceil \frac{x}{2} \right\rceil + \left\lfloor \frac{x}{2} \right\rfloor = \begin{cases} x & \text{if } x \text{ is even} \\ x+1 & \text{if } x \text{ is odd} \end{cases} \quad (8)$$

$$\underbrace{a_1 + a_2 + \cdots + a_n}_{\text{finite series}} = \sum_{r=1}^n a_r \quad (9)$$

8 Calculus and Vector Groups

$$\frac{\partial^2 f}{\partial x \partial y} = \nabla \cdot \vec{F} + \nabla \times \vec{G} \quad (10)$$

$$\oint_C (P dx + Q dy) = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dA \quad (11)$$

$$\lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = f'(x) \quad (12)$$

9 Matrices

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix} \quad (13)$$

$$B = \begin{bmatrix} \frac{1}{2} & -x & \sqrt{y} \\ a_1 & a_2 & a_n \\ 0 & 1 & \infty \end{bmatrix} \quad (14)$$

$$\det \begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc \quad (15)$$

10 Cases

$$|x| = \begin{cases} x & \text{if } x > 0 \\ 0 & \text{if } x = 0 \\ -x & \text{if } x < 0 \end{cases} \quad (16)$$

$$f(x) = \begin{cases} x^2 & \text{if } x > 1 \\ \frac{1}{1+x^2} & \text{if } -1 \leq x \leq 1 \\ x^3 & \text{if } x < -1 \end{cases} \quad (17)$$

$$h(t) = \begin{cases} \sqrt{t} & \text{if } t \geq 0 \\ \sqrt{-t} & \text{if } t < 0 \end{cases} \quad \begin{cases} 1 & \text{for a true proposition} \\ 0 & \text{for a false proposition} \end{cases} \quad (18)$$

11 Math Entry Modes and Token Boundaries

Dollar-delimited $x + 1$, parenthesis-delimited $x + 1$, adjacent ab , and punctuation after inline formulas: x_i , y^2 ; $z = 0$. An escaped dollar outside math costs \$5; a following formula is $5 + 2 = 7$.

$$x + 1 = 2$$

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Token boundaries: x^12 , x^{12} , a_12 , a_{12} , $\frac{1}{2}$, $\frac{12}{34}$, $\sqrt{2}$, and $\sqrt{23}$.

$$a + b = c$$

12 Fraction Styles and Continued Fractions

Inline fraction sizes: $\frac{a}{b}$, $\frac{a}{b}$, $\frac{a}{b}$. Inline binomial sizes: $\binom{n}{k}$, $\binom{n}{k}$, $\binom{n}{k}$.

$$\frac{a+b}{c+d} \quad \frac{a+b}{c+d} \quad \frac{a+b}{c+d}$$

$$\binom{n+1}{k+1} \quad \binom{n+1}{k+1} \quad \binom{n+1}{k+1}$$

$$\frac{1}{1 + \frac{1}{2 + \frac{1}{3 + x}}}$$

$$\frac{\frac{a}{b} + \frac{c}{d}}{\frac{e}{f} - \frac{g}{h}} \quad \frac{1}{\frac{1}{x}}$$

$$\binom{n}{k} \quad \left[\frac{a+b}{c+d} \right]$$

$$\frac{a}{b} \quad \frac{a}{b}$$

Plain TeX fractions stay inside their own groups:

$$\frac{a+b}{c+d} \qquad \binom{n}{k} \qquad \frac{a}{b}$$

$$\frac{a}{b} \qquad \left\{ \begin{matrix} a \\ b \end{matrix} \right\} \qquad \left[\begin{matrix} a \\ b \end{matrix} \right]$$

13 Indexed and Nested Radicals

Inline roots: $\sqrt{x}$, $\sqrt[3]{x}$, $\sqrt[n+1]{x^n+y^n}$.

$$\sqrt{\sqrt{x}} \qquad \sqrt[3]{\sqrt[4]{x}} \qquad \sqrt{1+\sqrt{1+\sqrt{1+x}}}$$

$$\sqrt{\frac{a^2+b^2}{c^2+d^2}} \qquad \frac{\sqrt[3]{x^2}}{\sqrt{1+x^2}}$$

$$\sqrt[\frac{n+1}{2}]{\left(\frac{a}{b}\right)^{n+1}}$$

14 Scripts, Primes, and Empty Groups

Script order: a_i^n , a_i^n , a_{ij}^{n+1} , a_{ij}^{n+1} . Primes: $f'(x)$, $f''(x)$, $f'''(x)$, $f'(x)$. Empty bases and groups: ${}_i^jX$, ${}_6^{14}\text{C}$, ab , x .

$$x_{ij}^{2^k} \qquad x_{ij_k}^{a^{b^c}} \qquad (x_i^2)_j^{n+1}$$

$$\frac{x_i^2+y_j^3}{z_{k+1}^{n-1}} \qquad \sqrt{x_{ij}^{2^k}+y^2}$$

$$\left(\frac{a}{b}\right)_{i,j}^{n+1} \qquad \sum_{i=1}^{n^2} x_{i_j}^{k+1}$$

15 Complete Greek Alphabet and Variants

$\alpha, \beta, \gamma, \delta, \epsilon, \zeta, \eta, \theta$

$\iota, \kappa, \lambda, \mu, \nu, \xi, \pi, \rho$

$\sigma, \tau, \upsilon, \phi, \chi, \psi, \omega$

$\varepsilon, \vartheta, \varpi, \varrho, \varsigma, \varphi$

$\Gamma, \Delta, \Theta, \Lambda, \Xi, \Pi, \Sigma, \Upsilon, \Phi, \Psi, \Omega$

16 Binary Operators and Unary Spacing

$a \pm b, \quad a \mp b, \quad a \times b, \quad a \div b, \quad a \cdot b$

$a * b, \quad a \star b, \quad a \circ b, \quad a \bullet b$

$a \oplus b, \quad a \ominus b, \quad a \otimes b, \quad a \oslash b, \quad a \odot b$

$A \cap B, \quad A \cup B, \quad A \sqcap B, \quad A \sqcup B, \quad A \setminus B$

$p \vee q, \quad p \wedge q, \quad p \vee q, \quad p \wedge q, \quad A \wr B, \quad A \amalg B$

$a \dagger b, \quad a \ddagger b, \quad a \boxplus b, \quad a \boxminus b$

$a \boxtimes b, \quad a \boxdiv b$

$-x, \quad +x, \quad a + -b, \quad a - (-b), \quad a = (-b), \quad (-a) + b$

$a + b = c, \quad a + b = c, \quad a(+b) = c, \quad a, b; c : d$

17 Relations, Logic, and Negation

$a \leq b, \quad a \geq b, \quad a \neq b, \quad a \equiv b, \quad a \approx b$

$a \sim b, \quad a \simeq b, \quad a \cong b, \quad a \asymp b, \quad a \doteq b$

$a \prec b, \quad a \succ b, \quad a \preceq b, \quad a \succeq b$

$A \subset B, \quad A \supset B, \quad A \subseteq B, \quad A \supseteq B$

$A \sqsubseteq B, \quad A \sqsupseteq B, \quad x \in A, \quad A \ni x, \quad x \notin B$

$p \vdash q, \quad p \dashv q, \quad p \models q, \quad a \mid b, \quad a \nmid b$

$$\begin{aligned}
& a \parallel b, \quad a \perp b, \quad a \propto b, \quad a \bowtie b, \quad a \ll b, \quad a \gg b \\
& A \trianglelefteq B, \quad A \trianglerighteq B, \quad a \not\leq b, \quad a \not\geq b \\
& a \not\leq b, \quad A \nVdash B \\
& a \neq b, \quad a \not\prec b, \quad a \not\succ b, \quad A \not\subset B \\
& \forall x \in A, \quad \exists y \in B, \quad \nexists z \in \emptyset, \quad \neg p \iff \neg p
\end{aligned}$$

18 Arrows and Labeled Relations

$$\begin{aligned}
& A \leftarrow B, \quad A \rightarrow B, \quad A \leftrightarrow B \\
& A \Leftarrow B, \quad A \Rightarrow B, \quad A \Leftrightarrow B \\
& A \longleftarrow B, \quad A \longrightarrow B, \quad A \longleftrightarrow B \\
& A \Longleftarrow B, \quad A \Longrightarrow B, \quad A \Longleftrightarrow B \\
& \uparrow, \downarrow, \Uparrow, \Downarrow, \Updownarrow, \nearrow, \nwarrow, \searrow, \swarrow \\
& A \hookrightarrow B, \quad A \leftarrow B, \quad x \mapsto x^2, \quad x \longmapsto x^3 \\
& A \twoheadrightarrow B, \quad A \leftarrow B, \quad A \rightleftarrows B, \quad A \rightrightarrows B \\
& A \leftharpoonup B, \quad A \rightarrowtail B, \quad A \lhd B, \quad A \rhd B \\
& A \curlywedge B, \quad A \frown B, \quad A \rightsquigarrow B \\
& A \ltimes B, \quad A \oslash B, \quad A \rightleftarrows B \\
& A \rightleftharpoons B, \quad A \Rightarrow B, \quad A \Leftarrow B, \quad A \Rightarrow B \\
& \Uparrow, \quad \Downarrow, \quad A \leftarrow B, \quad p \implies q \\
& A \xrightarrow{f} B \quad A \xleftarrow{g} B \\
& A \xrightarrow[h]{\frac{f+g}{2}} B \quad A \xleftarrow[n \rightarrow \infty]{\text{inverse}} B
\end{aligned}$$

19 Large Operators and Explicit Limits

Inline limits: $\sum_{i=1}^n x_i$, $\sum_{i=1}^n x_i$, $\int_0^1 f(x) dx$, and $\int_0^1 f(x) dx$.

$$\begin{array}{cccc}
 \sum_{i=1}^n x_i & \sum_{i=1}^n x_i & \prod_{i=1}^n x_i & \prod_{i=1}^n X_i \\
 \bigcup_{i=1}^n A_i & \bigcap_{i=1}^n A_i & \bigsqcup_{i=1}^n A_i & \biguplus_{i=1}^n A_i \\
 \bigvee_{i=1}^n p_i & \bigwedge_{i=1}^n p_i & \bigoplus_{i=1}^n V_i & \\
 \bigotimes_{i=1}^n V_i & \bigodot_{i=1}^n V_i & & \\
 \int_0^1 f(x) dx & \iint_D f dA & \iiint_V f dV & \oint_C f ds \\
 \liminf_{n \rightarrow \infty} a_n \leq \limsup_{n \rightarrow \infty} a_n & & & \\
 \sup_{x \in A} f(x) & \inf_{x \in A} f(x) & \max_{1 \leq i \leq n} x_i & \min_{1 \leq i \leq n} x_i \\
 \operatorname{Res}_{z=0} \frac{e^z}{z} & \operatorname{Res}_{z=0} \frac{e^z}{z} & &
 \end{array}$$

20 Named and Custom Functions

$$\begin{array}{l}
 \arcsin x, \quad \arccos x, \quad \arctan x, \quad \sec x, \quad \csc x, \quad \cot x \\
 \sinh x, \quad \cosh x, \quad \tanh x, \quad \coth x \\
 \arg z, \quad \deg p, \quad \dim V, \quad \ker T, \quad \operatorname{hom}(A, B) \\
 \gcd(a, b), \quad \lg n, \quad \Pr(A \mid B), \quad \log_2 n \\
 \operatorname{rank}(A) = \operatorname{tr}(P) \quad \operatorname{sgn}(x) |x| = x
 \end{array}$$

21 Math Alphabets and Text Styles

Abc123 **Abc123** *Abc123*

Abc123 Abc123 RNCQZ

ABEFHILMR *A B E F H I L M R* *ℰℋℐℜℤabc*

$$\mathbf{x}_{\max} \in \mathbb{R}^{n \times m} \quad \mathcal{L}\{f\}(s) = \int_0^\infty e^{-st} f(t) dt$$

$$e^{i\pi} + 1 = 0 \quad \alpha + \beta = \gamma$$

A*b***C** AbC $\mathbb{R} \times \mathbb{C} \times \mathbb{Z}$

Text spacing: a b versus ab, and a b .

plain words roman words **bold words**

italic words sans words mono words

boxed text $x_{\text{initial value}}^{\text{final value}}$

price: \$5; rate: 50%; key: a_b; A&B

22 Display, Text, and Script Styles

Inline comparison: $\sum_{i=1}^n \frac{1}{i}$ versus $\sum_{i=1}^n \frac{1}{i}$.

$$\frac{a}{b} \quad \frac{a}{b} \quad \frac{a}{b} \quad \frac{a}{b}$$

$$\sum_{i=1}^n x_i \quad \sum_{i=1}^n x_i \quad \sum_{i=1}^n x_i \quad \sum_{i=1}^n x_i$$

$$\frac{1 + \frac{a}{b}}{1 + \frac{c}{d}} \quad x_i^2 + x_i^2$$

23 Delimiter Families, Sizing, and Middles

$$\begin{array}{c}
 \left\{ \frac{a}{b} \right\} \quad \left| \frac{a}{b} \right| \quad \left\| \frac{a}{b} \right\| \\
 \left[\frac{a}{b} \right] \quad \left\lceil \frac{a}{b} \right\rceil \quad \left\langle \frac{a}{b} \right\rangle \\
 \frac{d}{dx} x^n \Big|_{x=1} \quad \left\{ \frac{a}{b} \right. \\
 \left\{ x \in \mathbb{R} \left| \frac{x}{1+x^2} > 0 \right. \right\} \\
 \left(\frac{a}{b} \middle| \frac{c}{d} \middle| \frac{e}{f} \right) \\
 \left[\left(\frac{1}{1 + \left| \frac{x}{y} \right|} \right)^2 \right]^{1/2} \\
 (x) \quad (x) \quad (x) \quad (x) \quad (x) \\
 [x] \quad [x] \quad [x] \quad [x] \\
 \{a \mid b\} \quad \{a \mid b\} \quad \{a \mid b\} \quad \left\{ a \middle| b \right\} \\
 \left| \frac{x}{y} \right| \quad \left\| \frac{x}{y} \right\| \quad \left(0, \frac{1}{2} \right] \\
 \uparrow \frac{a}{b} \downarrow \quad / \frac{a}{b} \backslash
 \end{array}$$

24 Accents and Stacked Annotations

$$\begin{array}{c}
 \hat{x}, \quad \check{x}, \quad \tilde{x}, \quad \acute{x}, \quad \grave{x}, \quad \breve{x} \\
 \bar{x}, \quad \vec{x}, \quad \dot{x}, \quad \ddot{x}, \quad \ddot{x}, \quad \ddot{x}, \quad \mathring{x} \\
 \widehat{ABC} \quad \widetilde{abcdefgh} \quad \overline{a+b} \quad \underline{a+b}
 \end{array}$$

$$\begin{array}{ccccccc} \hat{\hat{x}} & \dot{\dot{x}} & \overline{\overline{a}} & \widehat{\widehat{x_i^2+y_j^2}} \\ \overrightarrow{\overrightarrow{AB}} & \overleftarrow{\overleftarrow{AB}} & \overleftrightarrow{\overleftrightarrow{AB}} & \\ \underline{\underline{AB}} & \overleftarrow{\overleftarrow{AB}} & \overleftrightarrow{\overleftrightarrow{AB}} & \\ \overbrace{a+b+\cdots+z}^{n\text{ terms}} & & \underbrace{a+b+\cdots+z}_{n\text{ terms}} & \\ & \overbrace{a+b+c+d}^4 & & \\ & \underbrace{\hspace{1.5cm}}_2 & & \\ \stackrel{\text{def}}{=} & \lim_{n\rightarrow\infty} & \xrightarrow{f} & \\ & \overset{\alpha}{X}_{\beta} & \xrightarrow{\frac{1}{n}} & \end{array}$$

25 Additional Matrices and Arrays

$$\begin{array}{ccc} a & b & \\ c & d & \end{array} \qquad \left\{ \begin{array}{cc} a & b \\ c & d \end{array} \right\} \qquad \left\| \begin{array}{cc} a & b \\ c & d \end{array} \right\|$$

Compact inline matrix: $(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix})$.

$$(x) \qquad [a \quad b \quad c] \qquad \begin{pmatrix} a \\ b \\ c \end{pmatrix}$$

$$\begin{array}{ccc} 1 & 22 & 333 \\ 4444 & 55 & 6 \\ -x & \frac{1}{2} & \sqrt{y} \end{array}$$

$$\begin{array}{ccc} a & c & a \ b \\ & b & c \\ c & a & \end{array}$$

$$\begin{bmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{n1} & \cdots & a_{nn} \end{bmatrix}$$

$$\begin{pmatrix} a & b & 0 \\ c & d & \\ 0 & e & f \\ & g & h \end{pmatrix}$$

$$\frac{\det \begin{pmatrix} a & b \\ c & d \end{pmatrix}}{\sqrt{a^2 + b^2}} \quad \begin{pmatrix} a & b \\ c & d \end{pmatrix}_{ij}^2$$

26 Nested Cases and Aligned Groups

$$g(x) = \begin{cases} \begin{cases} x & \text{if } x < 0 \\ x^2 & \text{if } x \geq 0 \end{cases} & \text{if } |x| < 1 \\ \frac{1}{1+x^2} & \text{otherwise} \end{cases}$$

$$\left[\begin{array}{cc} \begin{cases} 0 & x < 0 \\ 1 & x \geq 0 \end{cases} & \frac{1}{2} \\ \sqrt{x^2 + 1} & \sum_{i=1}^n i \end{array} \right]$$

$$\begin{aligned} (a+b)^2 &= a^2 + 2ab + b^2 \\ (a-b)^2 &= a^2 - 2ab + b^2 \end{aligned} \tag{19}$$

$$\begin{aligned} (a+b+c)^2 &= a^2 + b^2 + c^2 \\ &\quad + 2ab + 2ac + 2bc \end{aligned} \tag{20}$$

27 Multiline Document Environments

$$x + y = 3 \tag{21}$$

$$2x - y = 0 \tag{22}$$

$$\begin{array}{ll} f(x) = x^2 & f'(x) = 2x \\ g(x) = e^x & g'(x) = e^x \end{array}$$

$$x = 1, \quad y = 2 \tag{23}$$

$$u = 3, \quad v = 4 \tag{24}$$

$$\begin{array}{ll} a + b = c, & a - b = d \\ 2a = c + d, & 2b = c - d \end{array}$$

$$a + b = b + a \tag{25}$$

$$(a + b) + c = a + (b + c) \tag{26}$$

$$\begin{array}{l} \sin^2 x + \cos^2 x = 1 \\ \cosh^2 x - \sinh^2 x = 1 \end{array}$$

$$\begin{array}{l} a + b + c + d + e \\ + f + g + h + i + j \\ + k + l + m + n = 0 \end{array} \tag{27}$$

$$\begin{array}{l} \frac{1}{1+x} + \frac{1}{1-x} \\ \\ = \frac{2}{1-x^2} \end{array}$$

28 Equation Tags, Labels, and References

$$a + b = b + a \tag{28}$$

$$a + 0 = a$$

$$a \cdot 1 = a \tag{Identity}$$

$$\begin{array}{l} x-x=0 \\ x+x=2x \end{array} \qquad \text{Zero}$$

The commutative law is equation (28).
The identity law is equation Identity.

29 Explicit Spaces, Kerns, and Rules

$$\begin{array}{ccccccccc} ab, & a\,b, & a\,b, & a\,b, & ab & & & & \\ ab, & a\,b, & a\,b, & a\,b, & ab & & & & \\ & a\,b & a\,b & a\,b & ab & & & & \\ & a\,b & a\,b & a\,b & ab & & & & \\ & ab & a\,b & ab & ab & & & & \\ \int_0^1 f(x)\,dx & & \int_0^1 f(x)dx & & dx & & dx & & \\ & & \text{---} & | & \blacksquare & & & & \end{array}$$

30 Invisible Boxes and Overlap

$$\begin{array}{ccccccc} a+ & +c & & a+b+c & & & \\ & \frac{1}{2} & & \frac{1}{+2} & & & \\ & \sqrt{x} & & \sqrt{x} & & & \\ & \sqrt{y} & & \sqrt{y} & & & \\ AB & & AB & & \lim_{n\rightarrow\infty} & & \end{array}$$

31 Boxes and Scoped Colors

$$\boxed{E = mc^2} \qquad \boxed{\frac{a+b}{c+d}} \qquad \boxed{\boxed{x} + y}$$

$$\textcolor{red}{x} + \textcolor{red}{y} + z \qquad x + \textcolor{blue}{y} + \textcolor{blue}{z} + w$$

$$\textcolor{red}{a} + \textcolor{blue}{b} + \textcolor{red}{c} + d$$

$$\frac{\textcolor{green}{\sqrt{x}}}{\textcolor{green}{1+x^2}} \qquad \textcolor{yellow}{x^2 + y^2}$$

$$\begin{pmatrix} \textcolor{red}{a} & b \\ c & \textcolor{blue}{d} \end{pmatrix} \qquad \boxed{\sum_{i=1}^n \textcolor{blue}{i}}$$

32 Miscellaneous Symbols and Ellipses

$$\ell, \wp, \Re, \Im, \aleph, \beth, \hbar, \iota, j$$

$$\emptyset, \varnothing, \top, \perp, \angle, \measuredangle, \triangle, \surd$$

$$\flat, \natural, \sharp, \clubsuit, \diamondsuit, \heartsuit, \spadesuit$$

$$\checkmark, \blacktimes, \blacktriangle, \blacksquare, \blacklozenge$$

$$\lrcorner, \mathbb{k}, \quad x: A \rightarrow B, \quad x \in \{a, b\}$$

$$|x|, \quad \|x\|, \quad 1, 2, \dots, n$$

$$a_1, \dots, a_n \qquad a_1 + \dots + a_n \qquad \vdots \qquad \ddots$$

$$\{x \mid x > 0\}, \quad \#1, \quad 50\%, \quad \$5, \quad a_b, \quad A\&B$$

33 Document Macros in Math

$$\mathbf{u} + \mathbf{v} = \mathbf{w} \qquad \langle \mathbf{u}, \mathbf{v} \rangle = \sum_{i=1}^n u_i v_i$$

$$\frac{dx^n}{dx} = nx^{n-1} \qquad \frac{dt^3}{dt} = 3t^2$$

$$\mathrm{rank}(A) = \dim \mathrm{im}(A)$$

$$\mathrm{rank}(A) + \dim \mathrm{ker}(A) = n$$

34 Deeply Nested Mixed Structures

$$\left\{\left\{\frac{\sum\limits_{i=1}^n x_{i_j}^{2^k}}{\sqrt[3]{1+\left(\frac{a+b}{c+d}\right)^2}}\right\}^{p+1}\right\}_m$$

$$\left\langle \widehat{\frac{\mathbf{u}}{\sqrt{1+u_i^2}},\frac{\mathbf{v}}{\sqrt[3]{1+v_j^3}}}\right\rangle$$

$$\frac{\overbrace{a_1+\cdots+a_n}^{n\text{ terms}}}{\underbrace{\sqrt{x_1^2+\cdots+x_n^2}}_{\text{norm}}}$$

$$\left[\begin{array}{cc} \frac{\sum_{i=1}^n i}{\sqrt{1+x^2}} & \xrightarrow{\alpha} \\ \left\{\begin{array}{ll} \mathbf{x} & x\geq 0 \\ -\mathbf{x} & x<0 \end{array}\right. & \sqrt[3]{\frac{1}{1+\frac{1}{1+x}}} \end{array}\right]^2$$

$$\frac{\left|\begin{array}{cc} a & b \\ c & d \end{array}\right|}{\left\|\begin{array}{c} x \\ y \end{array}\right\|}\overset[n\rightarrow\infty]{\text{normalized}}\mathbb{R}$$